

EDUCATION

Seoul National University (SNU)

Master of Science in Cognitive Science
Bachelor of Science in Mechanical Engineering
Minor in Philosophy

Seoul, South Korea

09/2026 – Present

08/2026

TOEFL iBT : 105 / 120

GPA : 3.76 / 4.00

PUBLICATIONS

The Limits of Your World Model Mean the Limits of Your Language Model: Embodied Reasoning through a Predictive Action Boundary

Jihun Moon*, Yumin Park*, Hogun Park*, Youngjae Yu (*equal contribution)

[Paper Link](#)

Under Review

Submitted 08/2026

SPREAD: Scalable Pre-trained World Model for Adaptive Dynamics Model

Jihun Moon, Seong-Woo Kim

[Paper Link](#)

IEEE Robotics and Automation Letters (RA-L), 2026

Accepted 04/2026

Beyond Destinations: Instruction-Aware Graph Path Planning and Navigation with OpenStreetMap

Donghwi Jung, Jihun Moon, Jiyang Lee, DongYeop Shin, Jungjae Lee, Jinhee Kim, Seong-Woo Kim

[Paper Link](#)

Under Review at RA-L 2026

Submitted 02/2026

RESEARCH EXPERIENCE

Physical Intelligence Lab, SNU

Seoul, South Korea

Undergraduate Researcher → M.S. Student

04/2026 – Present

- **Embodied Reasoning with World Models:** Co-first-authored EGO, grounding a VLM's action selection in a Top-K action boundary predicted by a V-JEPA 2-based world model on egocentric video (under review, 2026)
- **Open-Source Reproducibility Audit:** Identified and quantified a temporal-anchoring discrepancy in the released EK-100 anticipation evaluation of Meta's V-JEPA 2; proposed an upstream fix with a reproducible one-line experiment
- **Long-Term Egocentric World Models:** Investigating how world models should organize long-term experience, spanning event segmentation, temporal abstraction, and structured memory

Autonomous Robot Intelligence Laboratory, SNU

Seoul, South Korea

Undergraduate Researcher

05/2025 – 03/2026

- **World Model Continual Learning:** Proposed an online continual-learning method enabling pre-trained world models to adapt in real time to environmental changes (RA-L 2026)
- **Language-Guided Navigation:** Co-developed an LLM/VLM planning framework that extracts destinations, route constraints, and avoidances from user instructions (RA-L, under review)

EXTRACURRICULAR ACTIVITIES

Humanoid & Physical AI Club 'SHAPE'

Founding President

SNU

11/2025 – 07/2026

- **Organization Founding:** Established Seoul National University's first student organization dedicated to humanoid robotics and physical AI, and led it as founding president from launch through its first competition season
- **2026 Humanoid Challenge:** Led the club team's entry in the 2026 Humanoid Challenge, operating a humanoid platform with emphasis on autonomous task execution from natural language instructions
- **Club Management:** Ran recruitment, project planning, and team operations to build and sustain a multidisciplinary student research community

Robotics Startup Competitions

SNU · KAIST

Team Member

2026

- Developed and pitched robotics-based business ideas at the SNU GrandQuest Competition and the K-Robotics Startup Cup hosted by KAIST

MILITARY SERVICE

Republic of Korea Army, Joint Security Area Battalion (JSA, United Nations Command)

Paju, South Korea

Sergeant

12/2021 – 06/2023

- Served at South Korea's northernmost front-line outpost, conducting surveillance and security operations near the North Korean border

SCHOLARSHIPS

- SNU Merit-based & Additional Scholarship for Academic Excellence Awarded 08/2024
- SNU Merit-based Scholarship for Academic Excellence Awarded 02/2025
- SNU Merit-based Scholarship for Academic Excellence Awarded 02/2026

SELECTED PROJECTS

Maze Solving Manipulation

Project for Course 'Dynamics and Control of Robot-Environment Interaction'

10/2025 – 12/2025

- Implemented an A*-based path planner with a feedback trajectory tracking controller; achieved the top performance among all teams

Automatic Book Scanner

11/2024 – 12/2024

Project for Course 'Mechatronics'

- Designed and fabricated a multi-motor control system for automatic book scanner capable of page turning and synchronized screen transitions using a film suction pad mechanism

SKILLS

Machine Learning & AI

- **PyTorch:** Built world-model training pipelines; LoRA-based post-training of VLMs (Qwen3-VL) on frozen V-JEPA 2 representations
- **LLM/VLM:** Used LLMs and VLMs for instruction parsing, constraint extraction, and prompt design

Robot Control & Dynamics

- **Model-based Control:** Implemented model predictive control integrated with learned world model
- **Control Theory:** Completed graduate course "Dynamics and Control of Robot-Environment Interaction"

ROS & MuJoCo

- **ROS 2:** Conducted high-level path planning and navigation for ROS-based quadruped robot project
- **MuJoCo:** Utilized control theory to MuJoCo-based manipulator control and motion planning

Programming

- **Python, Linux:** Used extensively in Python for both team and independent research projects